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# SPECIFICATION

## Propylene Glycol Ph. Eur.* / USP* / JP* parenteral grade
**Propylene Glycol USP/EP** **product code: 220**

Testing specifications: Ph. Eur.* / USP* / JP* / LSM 029*/ LSM 220*
**The material meets all requirements of Ph. Eur.*, USP*, JP*, FCC* and (EU) No 231/2012 (E1520)**

| Parameter | Ph. Eur.* | | USP* | | JP* | |
|---|---|---|---|---|---|---|
| | Specification | Method | Specification | Method | Specification | Method |
| Assay | — | — | ≥ 99.5% | GC, LSM 029*/** | — | — |
| Characters / Description | conforms | Ph. Eur.* | — | — | conforms | JP* |
| Identification | D | Ph. Eur.* | A / B / C** | USP* | 1 / 2 | JP* |
| Distilling range | — | — | — | — | ≥ 95% (V/V) at 184-189°C | JP* |
| Appearance | clear, colourless | Ph. Eur.* | — | — | — | — |
| Acidity | ≤ 8.3 ml 0.01 N NaOH | Ph. Eur.* | ≤ 0.20 ml 0.10 N NaOH | USP* | conforms | JP* |
| Oxidising substances | ≤ 0.2 ml 0.05 N Sodium thiosulfate | Ph. Eur.* | — | — | — | — |
| Reducing substances | conforms | Ph. Eur.* | — | — | — | — |
| Chloride | — | — | ≤ 70 ppm | USP* | ≤ 0.007% | JP* |
| Sulfate | — | — | ≤ 60 ppm | USP* | ≤ 0.002% | JP* |
| Glycerin | — | — | — | — | conforms | JP* |
| Relative density (Ph. Eur.*)/ Specific gravity (USP*/JP*) | $d_{20}^{20}$: 1.035 – 1.040 | Ph. Eur.* | $d_{25}^{25}$: 1.035 – 1.037 | USP* | $d_{20}^{20}$: 1.035 – 1.040 | JP* |
| Refractive index | $n_D^{20}$: 1.431 – 1.433 | Ph. Eur.* | — | — | — | — |
| Residue on ignition | — | — | ≤ 3.5 mg / 50 g | USP* | ≤ 0.005% | JP* |
| Sulfated ash | ≤ 0.01% m/m | Ph. Eur.* | — | — | — | — |
| Water | ≤ 0.2% | Ph. Eur.* | ≤ 0.2% | USP* | ≤ 0.5% | JP* |
| Ethylene glycol and diethylene glycol (Ph. Eur.*) / Limit of diethylene glycol and ethylene glycol (USP*):<br>- DEG<br>- EG | <br><br><br>≤ 0.10%<br>≤ 620 ppm | GC, LSM 029*/** | <br><br><br>≤ 0.10%<br>≤ 0.10% | GC, LSM 029*/** | — | — |
| Ethylene glycol, diethylene glycol and related substances:<br>- Ethylene glycol<br>- Diethylene glycol<br>- ∑ Impurities<br>- each other related substance | — | — | — | — | <br>≤ 0.1%<br>≤ 0.1%<br>≤ 1.0%<br>≤ 0.1% | GC, LSM 029*/** |

**The requirements of Ph. Eur.* Ethylene glycol, diethylene glycol, of USP* Identification B/C, USP* Assay and of JP* Ethylene glycol, diethylene glycol and related substances are covered by internal validated GC method LSM 029*.

*current version

Effective date: 08.04.2025

Supersedes version from: 01.12.2024

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## Propylene Glycol Ph. Eur.* / USP* / JP* parenteral grade
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| Additional Specification | | |
|---|---|---|
| **Parameter** | **Specification** | **Test method** |
| Assay | ≥ 99.80% m/m | GC, LSM 029*/** |
| Monoethylene Glycol | ≤ 0.01% m/m | GC, LSM 029*/** |
| 1,2-Butanediol | ≤ 0.01% m/m | GC, LSM 029*/** |
| 1,3-Propanediol | ≤ 0.01% m/m | GC, LSM 029*/** |
| Diethylene Glycol | ≤ 0.01% m/m | GC, LSM 029*/** |
| Di- and Tripropylene Glycol | ≤ 0.1% m/m | GC, LSM 029*/** |
| ∑ Impurities | ≤ 1.0% (m/m) | GC, LSM 029*/** |
| Each other related substance | ≤ 0.1% (m/m) | GC, LSM 029*/** |

**The requirements of Ph. Eur.* Ethylene glycol, diethylene glycol, of USP* Identification B/C, USP* Assay and of JP* Ethylene glycol, diethylene glycol and related substances are covered by internal validated GC method LSM 029*.

| Microbiological Specification | | |
|---|---|---|
| **Parameter** | **Specification** | **Method** |
| Bioburden (∑ Total aerobic microbial count + Total yeast mould count) | ≤ 100 CFU/g | Ph. Eur.* 2.6.12 / USP* <61> / JP* (membrane filtration) |
| Endotoxins | ≤ 0.5 IU/mg | Ph. Eur.*/USP*/JP* (LAL, kinetic-turbidimetric method) |

| Quality Assurance and Quality Control | |
|---|---|
| **Process/Operation** | **Standard / Requirement** |
| Production | IPEC PQG – GMP Guidelines for Pharmaceutical Excipients* |
| Supply chain | IPEC PQG – GMP Guidelines for Pharmaceutical Excipients* & IPEC GDP Guideline* |
| Analytical quality control | Full analysis of specification for each batch in a GMP qualified laboratory (in compliance with EU-GMP Part I, Chapter 6*) |
| Batch release | Qualified Person according to EU-GMP (2001/83/EC; Articles 48 & 49) |
| Packaging | Grade D (100,000) clean room according to GMP [EU-GMP Part I Annex 1 (classification and operating conditions), US cGMP] |
| Packaging | Extended supervision of repackaging process |
| Primary packaging material | Specified packaging materials from audited suppliers:<br>1 L, 2.5 L Amber Glass Bottles<br>5 L, 10 L, 20 L HDPE Containers<br>200 L HDPE-Steel Combi-drums<br>1000 L HDPE EVOH Intermediate Bulk Container with diffusion barrier |
| Tamper evident sealing | Tamper evident closure with void label |
| Secure labelling | Full reconciliation of batch-specific labels |
| End-control of filled containers | First and last container of filling process (water content, colour, IR-spectrum) |

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**Storage:** It is recommended to store the material at ambient temperatures in closed containers and protected from sunlight and other sources of UV light.

**Shelf life:** 24 months from date of manufacture
The specified shelf-life relates to originally closed containers.

**Manufacturer and manufacturing site**: Dow Deutschland, Stade (Germany)

**Manufacturing process:** synthetic, key raw material: propylene oxide

### ***Regulatory Compliance:***

The material is stored, repacked, tested and released at Aug. Hedinger GmbH & Co. KG according to IPEC PQG – GMP Guidelines for Pharmaceutical Excipients* and EU-GMP Part II Guidelines*.

Regulatory information for this product, e.g., information on Aflatoxins, Allergens, BSE/TSE, Dietary Laws, Elemental Impurities (ICH Q3D*), GMO, Nitrosamines and Residual Solvents (ICH Q3C*) is available upon request.

**Batch certification:**

Every batch is analysed according to all parameters of this specification (except Description / Characters). The Certificate of Analysis (CoA) provides all test results including analysis date and date of manufacture. All CoAs are signed by a Qualified Person according to EU-GMP or a responsible QA/QC-Manager.

*current version

| Compiled by: 25.03.2025 | Approved by: 25.03.2025 | Released by: 02.04.2025 | Effective: | Supersedes: |
|---|---|---|---|---|
| Dr. Katja Teufel<br>QA-Manager | Dr. Stefan Heß<br>QA-Manager | Elisabeth Bartel<br>Qualified Person (GMP) | 08.04.2025 | 01.12.2024 |